Gaussian Elimination Method

Gaussian Elimination Method

System of Linear Equations:

$$\mathbf{Ax} = \mathbf{b}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{Bmatrix}$$

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Forward Elimination:

Gaussian Elimination Method

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$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a'_{22} & a'_{23} & \cdots & a'_{2n} \\ 0 & a'_{23} & a'_{33} & \cdots & a'_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & a'_{n2} & a'_{n3} & \cdots & a'_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b'_3 \\ \vdots \\ b'_n \end{Bmatrix}$$

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Forward Elimination:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a'_{22} & a'_{23} & \cdots & a'_{2n} \\ 0 & a'_{23} & a'_{33} & \cdots & a'_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & a'_{n2} & a'_{n3} & \cdots & a'_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b'_3 \\ \vdots \\ b'_n \end{Bmatrix} \qquad \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a'_{22} & a'_{23} & \cdots & a'_{2n} \\ 0 & 0 & a''_{33} & \cdots & a''_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & a''_{n3} & \cdots & a''_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b''_3 \\ \vdots \\ b''_n \end{Bmatrix}$$

Gaussian Elimination Method

System of Linear Equations:

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$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{Bmatrix}$$

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$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a'_{22} & a'_{23} & \cdots & a'_{2n} \\ 0 & a'_{23} & a'_{33} & \cdots & a'_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & a'_{n2} & a'_{n3} & \cdots & a'_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b'_3 \\ \vdots \\ b'_n \end{Bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a'_{22} & a'_{23} & \cdots & a'_{2n} \\ 0 & 0 & a''_{33} & \cdots & a''_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & a''_{n3} & \cdots & a''_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b''_3 \\ \vdots \\ b''_n \end{Bmatrix}$$

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$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a'_{22} & a'_{23} & \cdots & a'_{2n} \\ 0 & a'_{23} & a'_{33} & \cdots & a'_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & a'_{n2} & a'_{n3} & \cdots & a'_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b'_3 \\ \vdots \\ b'_n \end{Bmatrix}$$

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Back Substitution:

$$x_n = \frac{b_n^{(n-1)}}{a_{nn}^{(n-1)}}$$

Gaussian Elimination Method

System of Linear Equations:

$$\mathbf{Ax} = \mathbf{b}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{Bmatrix}$$

Forward Elimination:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a'_{22} & a'_{23} & \cdots & a'_{2n} \\ 0 & a'_{23} & a'_{33} & \cdots & a'_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & a'_{n2} & a'_{n3} & \cdots & a'_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b'_3 \\ \vdots \\ b'_n \end{Bmatrix} \quad \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a'_{22} & a'_{23} & \cdots & a'_{2n} \\ 0 & 0 & a''_{33} & \cdots & a''_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & a''_{n3} & \cdots & a''_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b''_3 \\ \vdots \\ b''_n \end{Bmatrix} \quad \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a'_{22} & a'_{23} & \cdots & a'_{2n} \\ 0 & 0 & a''_{33} & \cdots & a''_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a^{(n-1)}_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b''_3 \\ \vdots \\ b^{(n-1)}_n \end{Bmatrix}$$

Back Substitution:

$$x_n = \frac{b_n^{(n-1)}}{a_{nn}^{(n-1)}} \quad \rightarrow \quad x_i = \frac{b_i^{(i-1)} - \sum_{j=i+1}^{n} a_{ij}^{(i-1)} \cdot x_j}{a_{ii}^{(i-1)}}; \quad i = n-1, n-2, \dots, 1$$

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$$
\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\
E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\
E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}
$$

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$$\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\
E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\
E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}$$

Forward Elimination:

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$$\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
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E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\
E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}$$

Forward Elimination:

$E_2' = E_2 - (a_{21}/a_{11}) \times E_1$:

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$$\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\
E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\
E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}$$

Forward Elimination:

$$(\quad 2x_1 - 2x_2 + 3x_3 - 3x_4 = -20 \quad)$$

$$E_2' = E_2 - (a_{21}/a_{11}) \times E_1:$$

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$$
\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\
E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\
E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}
$$

Forward Elimination:

$$
\begin{array}{rlrcrcrcrcrl}
 & (& 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 &) \\
-\frac{2}{1}\times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &)
\end{array}
$$

$E_2' = E_2 - (a_{21}/a_{11}) \times E_1$:

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$$\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
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E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}$$

Forward Elimination:

$$\begin{array}{rcrcrcrcrcrl}
 & (& 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 &) \\
-\frac{2}{1}\times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\
\hline
E_2' = E_2 - (a_{21}/a_{11}) \times E_1: & & & & & - & x_3 & - & x_4 & = & -4 &
\end{array}$$

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$$\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\
E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\
E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}$$

Forward Elimination:

$$E_2' = E_2 - (a_{21}/a_{11}) \times E_1: \qquad \begin{array}{rrcrcrcrcrr}
 & (& 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 &) \\
-\frac{2}{1}\times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\
\hline
 & & & & & - & x_3 & - & x_4 & = & -4 & \\
\hline\hline
\end{array}$$

$$E_3' = E_3 - (a_{31}/a_{11}) \times E_1:$$

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$$\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\
E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\
E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}$$

Forward Elimination:

$$E_2' = E_2 - (a_{21}/a_{11}) \times E_1: \qquad \begin{array}{rrcrcrcrcrr}
& (& 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \;) \\
-\frac{2}{1}\times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \;) \\
\hline
& & & & & - & x_3 & - & x_4 & = & -4 \\
\hline\hline
\end{array}$$

$$E_3' = E_3 - (a_{31}/a_{11}) \times E_1: \qquad \begin{array}{rrcrcrcrcrr}
& (& x_1 & + & x_2 & + & x_3 & & & = & -2 \;)
\end{array}$$

Gaussian Elimination Method

<u>Example 03-01</u>: Find root of the following system of equations.

$$\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\
E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\
E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}$$

Forward Elimination:

$$E_2' = E_2 - (a_{21}/a_{11}) \times E_1: \quad \begin{array}{rlrcrcrcrcrl}
 & (& 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 &) \\
-\frac{2}{1} \times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\
\hline
 & & & & & - & x_3 & - & x_4 & = & -4 & \\
\hline\hline
\end{array}$$

$$E_3' = E_3 - (a_{31}/a_{11}) \times E_1: \quad \begin{array}{rlrcrcrcrcrl}
 & (& x_1 & + & x_2 & + & x_3 & & & = & -2 &) \\
-\frac{1}{1} \times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &)
\end{array}$$

Gaussian Elimination Method

<u>Example 03-01</u>: Find root of the following system of equations.

$$
\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\
E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\
E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}
$$

Forward Elimination:

$$
\begin{array}{rrrcrcrcrcrr}
 & (& 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 &) \\
-\frac{2}{1}\times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\
\hline
E_2' = E_2 - (a_{21}/a_{11}) \times E_1: & & & & & - & x_3 & - & x_4 & = & -4 &
\end{array}
$$

$$
\begin{array}{rrrcrcrcrcrr}
 & (& x_1 & + & x_2 & + & x_3 & & & = & -2 &) \\
-\frac{1}{1}\times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\
\hline
E_3' = E_3 - (a_{31}/a_{11}) \times E_1: & & & & 2x_2 & - & x_3 & + & x_4 & = & 6 &
\end{array}
$$

Gaussian Elimination Method

<u>Example 03-01</u>: Find root of the following system of equations.

$$
\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\
E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\
E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}
$$

Forward Elimination:

$$
E_2' = E_2 - (a_{21}/a_{11}) \times E_1: \quad
\begin{array}{rlrcrcrcrcrl}
 & (& 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 &) \\
-\frac{2}{1}\times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\
\hline
 & & & & & - & x_3 & - & x_4 & = & -4 &
\end{array}
$$

$$
E_3' = E_3 - (a_{31}/a_{11}) \times E_1: \quad
\begin{array}{rlrcrcrcrcrl}
 & (& x_1 & + & x_2 & + & x_3 & & & = & -2 &) \\
-\frac{1}{1}\times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\
\hline
 & & & & 2x_2 & - & x_3 & + & x_4 & = & 6 &
\end{array}
$$

$$
E_4' = E_4 - (a_{41}/a_{11}) \times E_1:
$$

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$$\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\
E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\
E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}$$

Forward Elimination:

$$E_2' = E_2 - (a_{21}/a_{11}) \times E_1: \qquad \begin{array}{r|rcrcrcrcr}
 & (\;2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20\;) \\
-\frac{2}{1}\times & (\;x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8\;) \\
\hline
 & & & & - & x_3 & - & x_4 & = & -4
\end{array}$$

$$E_3' = E_3 - (a_{31}/a_{11}) \times E_1: \qquad \begin{array}{r|rcrcrcrcr}
 & (\;x_1 & + & x_2 & + & x_3 & & & = & -2\;) \\
-\frac{1}{1}\times & (\;x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8\;) \\
\hline
 & & & 2x_2 & - & x_3 & + & x_4 & = & 6
\end{array}$$

$$E_4' = E_4 - (a_{41}/a_{11}) \times E_1: \qquad \begin{array}{rcrcrcrcr}
(\;x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4\;)
\end{array}$$

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$$\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\
E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\
E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}$$

Forward Elimination:

$$E_2' = E_2 - (a_{21}/a_{11}) \times E_1: \qquad \begin{array}{rrcrcrcrcrr}
 & (& 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \ ) \\
-\frac{2}{1}\times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \ ) \\
\hline
 & & & & & - & x_3 & - & x_4 & = & -4
\end{array}$$

$$E_3' = E_3 - (a_{31}/a_{11}) \times E_1: \qquad \begin{array}{rrcrcrcrcrr}
 & (& x_1 & + & x_2 & + & x_3 & & & = & -2 \ ) \\
-\frac{1}{1}\times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \ ) \\
\hline
 & & & & 2x_2 & - & x_3 & + & x_4 & = & 6
\end{array}$$

$$E_4' = E_4 - (a_{41}/a_{11}) \times E_1: \qquad \begin{array}{rrcrcrcrcrr}
 & (& x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4 \ ) \\
-\frac{1}{1}\times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \ )
\end{array}$$

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$$
\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\
E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\
E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4
\end{array}
$$

Forward Elimination:

$$
E_2' = E_2 - (a_{21}/a_{11}) \times E_1: \quad
\begin{array}{rrcrcrcrcrl}
& (\; 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 &) \\
-\frac{2}{1}\times & (\; x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\
\hline
& & & & - & x_3 & - & x_4 & = & -4 & \\
\hline\hline
\end{array}
$$

$$
E_3' = E_3 - (a_{31}/a_{11}) \times E_1: \quad
\begin{array}{rrcrcrcrcrl}
& (\; x_1 & + & x_2 & + & x_3 & & & = & -2 &) \\
-\frac{1}{1}\times & (\; x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\
\hline
& & & 2x_2 & - & x_3 & + & x_4 & = & 6 & \\
\hline\hline
\end{array}
$$

$$
E_4' = E_4 - (a_{41}/a_{11}) \times E_1: \quad
\begin{array}{rrcrcrcrcrl}
& (\; x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4 &) \\
-\frac{1}{1}\times & (\; x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\
\hline
& & & & & 2x_3 & + & 4x_4 & = & 12 & \\
\hline\hline
\end{array}
$$

Gaussian Elimination Method

Forward Elimination:

$$
\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2': & & & & - & x_3 & - & x_4 & = & -4 \\
E_3': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_4': & & & & & 2x_3 & + & 4x_4 & = & 12
\end{array}
$$

Gaussian Elimination Method

Forward Elimination:

$$
\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2': & & & & - & x_3 & - & x_4 & = & -4 \\
E_3': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_4': & & & & & 2x_3 & + & 4x_4 & = & 12
\end{array}
\rightarrow
\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\
E_4'': & & & & & 2x_3 & + & 4x_4 & = & 12
\end{array}
$$

Gaussian Elimination Method

Forward Elimination:

$$
\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2': & & & & - & x_3 & - & x_4 & = & -4 \\
E_3': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_4': & & & & & 2x_3 & + & 4x_4 & = & 12
\end{array}
\rightarrow
\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\
E_4'': & & & & & 2x_3 & + & 4x_4 & = & 12
\end{array}
$$

$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3''$:

Gaussian Elimination Method

Forward Elimination:

$$
\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2': & & & & - & x_3 & - & x_4 & = & -4 \\
E_3': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_4': & & & & & 2x_3 & + & 4x_4 & = & 12
\end{array}
\rightarrow
\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\
E_4'': & & & & & 2x_3 & + & 4x_4 & = & 12
\end{array}
$$

$$(\quad 2x_3 \quad + \quad 4x_4 \quad = \quad 12 \quad)$$

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'':$$

Gaussian Elimination Method

Forward Elimination:

$$
\begin{array}{lrrrrrrrrr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2': & & & & - & x_3 & - & x_4 & = & -4 \\
E_3': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_4': & & & & & 2x_3 & + & 4x_4 & = & 12
\end{array}
\rightarrow
\begin{array}{lrrrrrrrrr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2'': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\
E_4'': & & & & & 2x_3 & + & 4x_4 & = & 12
\end{array}
$$

$$
\begin{array}{rlrrrrrr}
 & (& & 2x_3 & + & 4x_4 & = & 12 \;) \\
-\frac{2}{-1} \times & (& - & x_3 & - & x_4 & = & -4 \;)
\end{array}
$$

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'':$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2': & & & & - & x_3 & - & x_4 & = & -4 \\
E_3': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_4': & & & & & 2x_3 & + & 4x_4 & = & 12
\end{array}
\rightarrow
\begin{array}{lrcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2'': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\
E_4'': & & & & & 2x_3 & + & 4x_4 & = & 12
\end{array}$$

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'': \qquad
\begin{array}{rl}
 & (\quad 2x_3 + 4x_4 = 12 \quad) \\
-\frac{2}{-1} \times & (\quad -x_3 - x_4 = -4 \quad) \\
\hline
 & \qquad\qquad 2x_4 = 4
\end{array}$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{llcrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & & - & x_3 & - & x_4 & = & -4 \\ E_3': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_4': & & & & & 2x_3 & + & 4x_4 & = & 12 \end{array} \rightarrow \begin{array}{llcrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\ E_4'': & & & & & 2x_3 & + & 4x_4 & = & 12 \end{array}$$

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'':$$

$$\begin{array}{rrcrcrcrr} & (& & 2x_3 & + & 4x_4 & = & 12 &) \\ -\frac{2}{-1} \times & (& - & x_3 & - & x_4 & = & -4 &) \\ \hline & & & & & 2x_4 & = & 4 & \end{array}$$

$$\begin{array}{llcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\ E_4''': & & & & & & & 2x_4 & = & 4 \end{array}$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{llcrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & & - & x_3 & - & x_4 & = & -4 \\ E_3': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_4': & & & & & 2x_3 & + & 4x_4 & = & 12 \end{array} \rightarrow \begin{array}{llcrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\ E_4'': & & & & & 2x_3 & + & 4x_4 & = & 12 \end{array}$$

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'':$$

$$\begin{array}{rrcrcrcrr} & (& & 2x_3 & + & 4x_4 & = & 12 &) \\ -\frac{2}{-1} \times & (& - & x_3 & - & x_4 & = & -4 &) \\ \hline & & & & & 2x_4 & = & 4 & \end{array}$$

$$\begin{array}{llcrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\ E_4''': & & & & & & & 2x_4 & = & 4 \end{array}$$

Back Substitution:

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{lrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & & - & x_3 & - & x_4 & = & -4 \\ E_3': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_4': & & & & & 2x_3 & + & 4x_4 & = & 12 \end{array} \rightarrow \begin{array}{lrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\ E_4'': & & & & & 2x_3 & + & 4x_4 & = & 12 \end{array}$$

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'': \qquad \begin{array}{r} (\quad 2x_3 + 4x_4 = 12 \quad) \\ -\frac{2}{-1} \times (\quad - x_3 - x_4 = -4 \quad) \\ \hline 2x_4 = 4 \end{array}$$

$$\begin{array}{lrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\ E_4''': & & & & & & & 2x_4 & = & 4 \end{array}$$

Back Substitution:

$$x_4 = \frac{4}{2} = 2$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{llcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2': & & & & - & x_3 & - & x_4 & = & -4 \\
E_3': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_4': & & & & & 2x_3 & + & 4x_4 & = & 12
\end{array}
\rightarrow
\begin{array}{llcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\
E_4'': & & & & & 2x_3 & + & 4x_4 & = & 12
\end{array}$$

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'': \qquad
\begin{array}{rr}
 & (\quad 2x_3 + 4x_4 = 12 \quad) \\
-\frac{2}{-1} \times & (\quad -x_3 - x_4 = -4 \quad) \\
\hline
 & 2x_4 = 4 \\
\hline\hline
\end{array}$$

$$\begin{array}{llcrcrcrcr}
E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\
E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\
E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\
E_4''': & & & & & & & 2x_4 & = & 4
\end{array}$$

Back Substitution:

$$x_4 = \frac{4}{2} = 2$$

$$x_3 = \frac{-4-(-x_4)}{-1} = \frac{-4-(-2)}{-1} = 2$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{lrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & & - & x_3 & - & x_4 & = & -4 \\ E_3': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_4': & & & & & 2x_3 & + & 4x_4 & = & 12 \end{array} \rightarrow \begin{array}{lrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\ E_4'': & & & & & 2x_3 & + & 4x_4 & = & 12 \end{array}$$

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'': \qquad \begin{array}{rl} & (\ 2x_3 + 4x_4 = 12\ ) \\ -\frac{2}{-1}\times & (\ -x_3 - x_4 = -4\ ) \\ \hline & 2x_4 = 4 \end{array}$$

$$\begin{array}{lrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\ E_4''': & & & & & & & 2x_4 & = & 4 \end{array}$$

Back Substitution:

$$x_4 = \frac{4}{2} = 2$$

$$x_3 = \frac{-4-(-x_4)}{-1} = \frac{-4-(-2)}{-1} = 2$$

$$x_2 = \frac{6-(-x_3)-x_4}{2} = \frac{6-(-2)-2}{2} = 3$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{lrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & & - & x_3 & - & x_4 & = & -4 \\ E_3': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_4': & & & & & 2x_3 & + & 4x_4 & = & 12 \end{array} \rightarrow \begin{array}{lrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\ E_4'': & & & & & 2x_3 & + & 4x_4 & = & 12 \end{array}$$

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'': \qquad \begin{array}{rrcrcrcrl} & (& & 2x_3 & + & 4x_4 & = & 12 &) \\ -\frac{2}{-1}\times & (& - & x_3 & - & x_4 & = & -4 &) \\ \hline & & & & & 2x_4 & = & 4 & \end{array}$$

$$\begin{array}{lrcrcrcrcr} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2': & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E_3'': & & & & - & x_3 & - & x_4 & = & -4 \\ E_4''': & & & & & & & 2x_4 & = & 4 \end{array}$$

Back Substitution:

$$x_4 = \frac{4}{2} = 2$$

$$x_3 = \frac{-4-(-x_4)}{-1} = \frac{-4-(-2)}{-1} = 2$$

$$x_2 = \frac{6-(-x_3)-x_4}{2} = \frac{6-(-2)-2}{2} = 3$$

$$x_1 = \frac{-8-(-x_2)-2x_3-(-x_4)}{1} = \frac{-8-(-3)-2\cdot(2)-(-2)}{1} = -7$$